COURSE TITLE : GENERAL CHEMISTRY
COURSE CODE : 3073
COURSE CATEGORY : B
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Inorganic Chemistry	10
2	Organic Chemistry	16
3	Organic Compounds containing Oxygen and Nitrogen	18
4	Physical chemistry	16
TOTAL		**60**

COURSE OUTCOMES:

SL.NO.	SUB	STUDENT WILL BE ABLE TO
1	1	To apply the laws of chemical combination.
	2	To describe the periodicity in properties
	3	To explain the shape of molecules
2	1	To understand the methods of purification of organic compounds.
	2	To comprehend the methods of qualitative and quantitative analysis of organic compounds.
	3	To elucidate the molecular formula of organic compounds.
3	1	To understand different types of classification of organic compounds
	2	To illustrate different types of isomerism
	3	To explain methods of preparation, nomenclature and uses of hydrocarbons
4	1	To illustrate methods of preparation,chemical properties and uses of aldehyde, ketones, carboxylic acid, ethers, aromatic amines and phenols
	2	To understand structure and chemical reactions of glucose
	3	To explain the structure and importance of aminoacids and proteins
5	1	To know the chemical equilibrium and different types ofequilibrium constant
	2	To know and apply Lechatlier’s principle to important industrial processes(Haber process and contact process)
	3	To understand the terms-rate.rate constant,order and molecularity of a chemical reaction. Explain effect of temperature on rate of a reaction
6	1	To know about behavior of gases using gas laws
	2	To state postulates of k inetic theory of gases
	3	To analyse behavior of a real gas using Van der waals equation

SPECIFIC OUTCOMES

MODULE I

1.1.0 Apply the laws of chemical combination.

1.1.1 Illustrate each of the following laws of chemical combination with two examples each: Law of conservation of mass, Law of definite proportions ,Law of multiple proportions, Law of reciprocal proportions, Gay lussac's law of gaseous volumes, Avagadro's law and mention its applications.

1.1.2 Solve problems based on the above laws of chemical combination

1.1.3 State Dulong and Petit's Law

1.1.4 Calculate atomic weight of an element from its equivalent weight

1.1.5 State the relationship between vapour density and molecular weight.

1.2.0 Narrate the periodicity in properties.

1.2.1 State modern periodic law and briefly discuss the modern periodic table. Explain the periodicity in the properties of elements in terms of the following: Atomic and ionic radii, Ionisation enthalpy, Electron gain enthalpy and electro negativity

1.2.2 Explain the concept of hybridization (sp^3,sp^2,sp) with one example each (methane ethene , ethyne)

MODULE II

2.1.0 Understand the classification of organic compounds

2.1.1 Classify the given organic compounds on the basis of functional group

2.1.2 Illustrate 'Homologous series with examples

2.1.3 Name any five homologous series

2.2.0 Understand the methods of purification of organic compounds.

2.2.1 State the criteria of purity of an organic compound

2.2.2 Describe the purification of organic solids by crystallization and sublimation with examples.

2.2.3 Describe the purification of organic liquids by distillation (Simple distillation, Fractional distillation, Distillation under reduced pressure and steam distillation) and chromatography(Column chromatography, Thin layer chromatography and paper chromatography)

2.3.0 Comprehend the methods of analysis of organic compounds

2.3.1 Describe any one test that can be used for detection of the following elements a) Hydrogenb) Carbon c)Nitrogen d)Halogens e)Sulphur f)Phosphorus

2.3.2 Describe the following experiments used for the estimation of the following elements: Hydrogen and Carbon by combustion method (Liebig' method) Nitrogen by Dumas method , Halogen by Carius method, Sulphur by Carius method

2.3.3 Solve problems based on the above methods of estimation

2.4.0 Elucidate the molecular formula of organic compounds

2.4.1 Define empirical formulae

2.4.2 Find the empirical formulae from the given percentage composition

2.4.3 Define the molecular formulae

2.4.4 Define structural formulae- dash formula, complete formula condensed formula and bond line formula

2.4.5 Describe the shape of the following molecules using hybridization a)Methane b)Ethane c)Ethene d) ethyne e)Benzene

2.5.0 Comprehend the different types of Isomerism

2.5.1 Name the first 5 members of alkane, alkene and alkyne series using I.U.P.A.C System

2.5.2 Define Isomerism

2.5.3 Illustrate the following classes of structural isomerism with an example for each

(a) Chain isomerism

(b)Functional group isomerism

(c)Position isomerism

(d) Metamerism

2.5.4 Illustrate stereoisomerism

a. Geometrical isomerism exhibited by Maleic acid and fumaric acid

b. Optical isomerism exhibited by lactic acid.

2.6.0 Understand the methods of preparation of Hydrocarbons

2.6.1 Describe one method for the preparation of the following hydrocarbons– Methane (decarboxylation method), ethane (Wurtz reaction) ,ethene(by dehydrohalogenation),Ethyne (action of water on calcium carbide) ,Benzene (cyclic polymerization of ethyne)

2.6.2 Describe chemical reaction of the following hydrocarbons methane,ethylene,Acetylene and benzene with chlorine under different conditions

2.6.3 List the uses of the following hydrocarbons methane, ethane, ethylene, Acetylene and Benzene.

2.6.4 Explain a) Addition reaction of bromine in CCl_4 and HBr to symmetrical and unsymmetrical alkenes –Markovnikov's rule and Anti Markovnikov's rule b)Acidity of ethyne -reaction with sodium and sodamide c) Friedel craft's alkylation of benzene.

MODULE III

3.1.1 Explain the methods of preparation, of ethyl alcohol (fermentation and using Grignard reagent),Esterification, Chlorination and oxidation reactions and uses of Ethylalcohol.

3.1.2 Explain the method of preparation of formaldehyde(from methanol) ,acetaldehyde(from ethanol) and acetone (from isopropyl alcohol),
Describe nucleophilic addition reaction (addition of HCN), Cannizaro's reaction and aldol condensation reaction of formaldehyde and acetaldehyde,oxidation reaction of acetaldehyde and acetone using strong and mild oxidising agents (test of aldehyde and ketone)

3.1.3 Explain method of preparation of acetic acid(from Grignard reagent) .Chemical properties of acetic acid –reduction and Hell-Volhard-Zelinsky (HVZ reaction)

3.1.4 Explain Williamson's Method for preparing Ether and to know the important precautions and uses of ether

3.1.5 Explain the method of preparation of aniline(from nitrobenzene) and diazotisation of aniline

3.1.6 Explain the method of conversion of aniline to phenol.chlorobenene and benzene

3.1.7 Describe the method of preparation of phenol by Dow's process and Cumene process. Electrophilic substituition reaction (nitration and bromination)and special reactions (-Kolbe reaction and Reimer-Tiemann reactions) of Phenol

3.1.8 Describe the structure of glucose (open chain and Ring structure)

3.1.9 Explain the chemical reactions of glucose –reduction,oxidation .acetylation and reaction with phenyl hydrazine

3.2.0 Comprehend the chemistry of Amino Acids and Proteins

3.2.1 Classification of aminoacids (essential and nonessential amino acid with 2 example each)

3.2.2 Explain the properties of amino acids(zwitter ion and isoelectric point) and concept of peptide bond

3.2.3 Describe the 1^0,$2,^0$ 3^0 structure of protein.

3.2.4 List the biological functions and test for proteins.

MODULE IV PHYSICAL CHEMISTRY

4.1.0 Understand the conditions of chemical equilibrium

4.1.1 Illustrate with example
1. Reversible 2. Irreversible reactions

4.1.2 State the law of mass action

4.1.3 Derive expression for the equilibrium constant K_p and Kc

4.1.4 Derive the relation between K_p and Kc and solve problems based on this.

4.1.5 Derive an expression for equilibrium constant for each of the following reactions
$H_2+I_2 = 2HI$
$N_2+3H_2=2NH_3$
$N_2O_4 = 2\ NO_2$

4.1.6 State Le-Chatlier's principle

4.1.7 Predict the optimum conditions for getting maximum yield of the product by Applying the Le-Chatlier's principle in each of the following process
(a) Manufacture of NH_3 by Haber process
(b) Manufacture of SO_3 by contact process

4.2.0 Understand the kinetics of chemical reactions

4.2.1 Define rate or velocity of a chemical reaction

4.2.2 Distinguish between order and molecularity of a chemical reaction

4.2.3 Illustrate first order reaction with two examples(acid catalyzed hydrolysis of ester and inversion of cane sugar)

4.2.4 Derive an equation for 1^{st} order reaction

4.2.5 Explain half-life period–derive an expression to determine the half-life period

4.2.6 Solve problems based on 1^{st} order reactions

4.2.7 Illustrate second order reactions with two examples (saponification reaction and H_2-I_2 reaction)

4.2.8 Illustrate third order reaction with one example: ($2NO_2+O_2 \rightarrow 2NO_3$)

4.2.9 Explain the effect of temperature on reaction rate and define the terms threshold energy . and activation energy ,state Arrhenius equation (exponential and logarithmic forms)

4.3.0 Comprehend the behaviour of Gaseous state

4.3.1 List the characteristics of Gaseous state and their physical properties- pressure, volume and temperature.

4.3.2 State Boyle's law, Charle's law and Avogadro's law and solve simple problems based on these laws.

4.3.3 Derive ideal gas equation (PV=nRT)

4.3.4 Know the significance of universal gas constant and calculate its value in different units.
4.3.5 State the postulates of kinetic theory
4.3.6 Distinguish between real gas and ideal gas and give the conditions and the reasons of deviations from ideal behaviour.
4.3.7 Explain the van der Walls equation of state of real gases and the significance of van der Waals constants.

CONTENT DETAILS

MODULE I: INORGANIC CHEMISTRY

LAWS OF CHEMICAL COMBINATION

Law of conservation of mass–Law of definite proportions–Law of multiple proportions -Law of reciprocal proportions–Gay Lussac's law of gaseous volumes - statement, examples and Problems based on the above laws.:

Avagadro's law- statement and applications- relationship between molecular weight and vapour density(No derivation)

Dulong and Petit's Law. statement, applications and simple problems based on this law. Give the relation to find out the atomic weight of element from its equivalent weight

PERIODICITY IN PROPERTIES

Statement of modern periodic law-explain the Moseley's modern periodic table with emphasis to periods and groups-explanation of periodicity of following physical properties along a period and down a group-atomic(covalent and Van der waals)radii and ionic radius .ionisation enthalpy, electron gain enthalpy and electro-negativity

SHAPE OF MOLECULES

Explanation of shape of molecules using the concept of hybridization - Definition and important characteristics of hybridization -Types of hybridization-SP^3.SP^2, SP- explanation of shape of organic molecules like methane, ethene and ethyne and inorganic molecules like water and ammonia using hybridization.

MODULE II: ORGANIC CHEMISTRY

CLASSIFICATION OF ORGANIC COMPOUNDS

Introduction to organic chemistry–functional groups-Homologous series-classification of organic compounds on the basis of functional group-Give any 5 examples of homologous series

PURIFICATION OF ORGANIC COMPOUNDS

Criteria of purity-melting point, boiling point and mixed melting point- Purification of solid organic compounds by crystallization and sublimation-Give the principle and two Examples

Purification of liquid organic compounds by distillation-principle and two examples of simple distillation, fractional distillation-distillation under reduced pressure and steam distillation

Chromatography- theory- Types of chromatography -Adsorption and partition chromatography - Adsorption chromatography- Column chromatography and thin layer chromatography

Partition chromatography-paper chromatography

QUALITATIVE AND QUANTITATIVE ANALYSIS OF ORGANIC COMPOUNDS

Give one method of detection of following elements along with the chemistry of reactions - carbon and hydrogen by combustion test , Nitrogen ,sulphur, and halogen by Lassaigne's test)and phosphorus by ammonium molybdate test .

Estimation of elements carbon and hydrogen by Liebig's method –Nitrogen by Dumas method-halogens by Carius method- Sulphur by Carius method-solve simple problems based on these methods of estimation.

MOLECULAR FORMULA OF ORGANIC COMPOUNDS

Definition of empirical formula-deduction of empirical formula from percentage composition of elements-definition of molecular formula, calculation of molecular formula from empirical formula-Structural formulae of compounds- dash formula, complete formula, condensed formula and bond line formula.

Shape of molecules using hybridization-methane and ethane by sp^3, ethene and benzene by sp^2 and ethyne by sp hybridization

PREPARATION OF HYDROCARBON

IUPAC names of first five members of alkanes alkenes and alkynes(straight chain and branched chain)

Preparation of hydrocarbons – methane by decarboxylation of carboxylic acid using sodalime distillation-ethane by Wurtz reaction of chloromethane - ethene by dehydrohalogenation of chloroethane using alcoholic KOH – ethyne by action of water on calcium carbide and benzene by cyclic polymerization of ethynes.

Chemical properties of hydrocarbons – chlorination of methane, ethane and ethyne, Benzene – substitution and addition reactions of chlorine under different conditions, Addition reaction of Br_2 in CCl_4 to alkenes (test of unsaturation) – Addition of HBr to symmetrical and unsymmetrical alkenes – Markovnikov's and AntiMarkovnikov's addition – Acidity of ehynes – reaction with sodium and sodamide (test of alkynes) – Friedel – Craft's methylation of benzene, Any three uses of methane, ethane, ethene, ethyne and benzene.

MODULE III: ORGANIC CHEMISTRY

Methods of preparation of ethanol - fermentation reaction and action of Grignard reagent on methanal – reaction of ethanol with ethanoic acid (esterification) – reaction of ethanol with chlorinating agents like Luca's reagent , PCl_3, PCl_5 and $SOCl_2$- oxidation reaction of ethanol using PCC and acidified $K_2Cr_2O_7$ or $KMnO_4$ – three uses of ethanol .

Preparation of methanal from methanol - ethnal from ethanol – proponone from

2-methyl propan-2-ol, Nucleophilic addition of HCN to methanal and ethanal- Cannizaro's reaction and aldol condensation reaction of methanal and ethanal – oxidation reactions of methanal, ethanal and proponone using strong oxidizing agents like alkaline or acidified $KMnO_4$ and mild oxidizing agents like Tollen's reagent and Fehling's solution (test of aldehydes and ketones).

Method of preparation of ethanoic acid from CO_2 using Grignard reagent – chemical properties – reduction reaction and HVZ reaction of ethanoic acid .

Preparation of symmetrical and unsymmetrical ethers - Williamson's synthesis, precaution of ether for surgical purpose – any three uses of ether.

Preparation of aniline- by reduction of nitrobenzene- diazotisation of aniline – formation of benzene diazonium chloride – conversion of benzene diazonium chloride to phenol , chlorobenzene (Sandmeyer's reaction and Gattermann reaction) and benzene.

Preparation of phenol – Dow's process (from chlorobenzene) and Cumene process (from benzene)- electrophilic substitution reaction of phenol – bromination (Br_2 in CS_2 and Br_2 in water) and nitration (dil.HNO_3 and Con.HNO_3) - separation of ortho and para nitro phenols – special reactions of phenol – Kolbe reaction and Reimer Tiemann reaction (no mechanism is required).

Structure of glucose – open chain and ring structure- chemical reactions of glucose – reduction using HI, oxidation using Br_2 in water and HNO_3 – acetylation using acetic anhydride and osazone formation using phenyl hydrazine .

Amino acid – give names of 5 alpha amino acids –examples of optically inactive aminoacid and aminoacid containing secondary amino group

classification of aminoacids- essential and non essential aminoacids with two examples each essential amino acids –valine and leucine, nonessential amino acids - glycine and alanine (structures are required) – properties of alpha amino acids- zwitter ion structure - definition and significance of isoelectric point – formation of peptide bond and proteins- structure of proteins - 1^0, 2^0 and 3^0 structures -.biological function of proteins- structural materials,transport agents and biological catalysts(enzyme) - test of proteins – biurett test ,xanthoproteic and ninhydrin test (no chemistry is required)

MODULE IV PHYSICAL CHEMISTRY

CHEMICAL EQUILIBRIUM

Reversible and Irreversible reactions with two examples for each-Statement of Law of mass action-Equilibrium constants-Equilibrium concentration of reactants and products-molar concentration and partial pressure-Derivation of expressions for equilibrium constants-K_c and Kp-Law of chemical equilibrium-Derivation of relation between K_c and Kp-Application of the relation to three chemical equilibria $H_2 + I_2 \leftrightarrow 2HI$, $N_2 + 3H_2 \leftrightarrow 2NH_3$ and $N_2O_4 \leftrightarrow 2NO_2$ and simple problems using this relation Lechateleir's principle-Statement and application of

Le – Chateleir's principle to synthesis of ammonia by Haber process and manufacture of sulphuric acid by contact process.

CHEMICAL KINETICS

Rate of chemical reaction- definition and unit-Factors affecting rate of chemical reaction -rate law and rate constant-order and molecularity –examples of first order, second order and third order reactions-examples of Unimolecular,bimolecular and termolecular reactions-Difference between order and molecularity of reaction- Derivation of an expression of rate constant of first order reactions-Definition of half life period and derivation of expression of half life period from expression of first order rate constant-problems using these expressions-Effect of temperature on reaction rate-Definition of Threshold energy and Activation energy- Arrhenius equation- Exponential forms and logarithmic forms- Explanation of terms- Arrhenius frequency factor and activation energy.

GASEOUS STATE

Characteristics of gaseous state- Properties of gases like pressure,volume and temperature –Behaviour of gases –Gas Laws –Boyles Law, Charles Law and Avagadro's law (Qualitative and Quantitative statements)- Problems based on these laws –Derivation of ideal gas equation of states –explanation of significance of universal gas constant –Derivation of values of universal gas constant in different units-Postulates of kinetic theory of gases –Deviation of behaviour of gas from ideal gas – conditions and reasons – Differences between ideal gas and real gas – Van Der Waal's equation of state of a real gas – Explanation of different terms and significance of Van Der Waal's constants along with their units.

REFERENCE

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P.L.Soni,O.P.Dharmatha	-	Textbook of physical Chemistry		
Samuel Glasstone	-	Textbook of physical Chemistry	-	Mc.GrawHill s publisher
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